Rohan Sai Jagan

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Bangalore, India

12-03-2004

PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience building modern, responsive web applications using React.js, Next.js, TypeScript, and REST APIs. Skilled in developing scalable UI components, integrating API-driven backend services, and optimizing frontend performance. Proficient with Tailwind CSS, Docker, Git, and modern frontend development practices, with a strong passion for building high-quality user experiences.

TECHNICAL SKILLS

Programming Languages
JavaScript (ES6+), TypeScript, Python, Java, SQL

Web & Backend Technologies
React.js, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design, FastAPI, REST APIs

Frameworks & Libraries
TensorFlow, OpenCV, Scikit-learn, NumPy, Pandas, LangGraph

AI & Machine Learning
Machine Learning, Deep Learning, Computer Vision, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Google Gemini API

Databases
MySQL, PostgreSQL, ChromaDB

Tools
Git, GitHub, Docker, Linux, Vite

Cloud & DevOps
AWS (EC2, S3, IAM), Docker, Git, GitHub, Linux (CLI)

Core Concepts
Component-Based Architecture, API-First Development, Microservices, UI/UX Principles, Cross-Browser Compatibility, Performance Optimization, SDLC

Core Computer Science
Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Development Life Cycle, System Design (Basics)

EDUCATION

Bachelor of Engineering in Computer Science
Sjb Institute of Technology
2023 – 2027 | Bangalore

Higher Secondary
Mahesh PU College (Allen Career Institute)
2020 – 2022 | Bangalore

Pre High School
Venkat International Public School
2019 – 2020 | Bangalore

PROJECTS

1. ComplianceX – AI-Powered Compliance Intelligence Platform
React.js | TypeScript | FastAPI | Python | REST APIs | Tailwind CSS | Docker
- Built a responsive compliance dashboard using React and reusable UI components.
- Integrated REST APIs with FastAPI for real-time compliance monitoring.
- Optimized frontend performance and responsive design across devices.

2. TerraGuard – Geospatial Digital Twin for Forest Resilience
Python | FastAPI | Scikit-learn | Leaflet.js | GIS | HTML/CSS | JavaScript
- Developed a geospatial platform for forest fire prediction using a Random Forest ML model and real-time weather APIs.
- Built interactive GIS dashboards and climate simulation for environmental monitoring and risk visualization.
- Developed REST APIs using FastAPI and integrated weather and geospatial services for real time predictions.
- Implemented environmental analytics and automated report generation for decision support.

3. Driver Drowsiness Detection System
Python | TensorFlow/Keras | OpenCV | CNN | NumPy
- Developed a real-time driver drowsiness detection system using a CNN and OpenCV.
- Implemented real-time fatigue detection through face and eye landmark detection, triggering alerts during prolonged eye closure.
- Trained and evaluated a custom CNN model, achieving 98.2% classification accuracy on the test dataset.
- Applied computer vision and deep learning techniques to improve road safety through real-time monitoring.

4. AI Document Intelligence Assistant
Python | FastAPI | Gemini API | ChromaDB | RAG | React
- Developing an AI-powered document assistant for semantic search, document summarization, and question answering over PDF documents.
- Implementing Retrieval-Augmented Generation (RAG) using ChromaDB and vector embeddings for context-aware responses.
- Building REST APIs with FastAPI for document ingestion, retrieval, and AI-powered document analysis.

ORGANISATIONS

Venkat International Public School
Event Management Lead
Led the planning and execution of school events by coordinating cross-functional teams, managing event logistics, and ensuring successful delivery, strengthening leadership, communication, and organizational skills.

SJBIT
Hackathon Volunteer & Event Coordinator
Volunteered in organizing a 24-hour hackathon, coordinating participant registrations, event logistics, and team support to ensure smooth execution of the event.